LERO Software for a better world

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CS5741
Concurrency & Parallelism in Software Development

Lecture 5 – Reflection

Dr Faeq Alrimawi

HOST INSTITUTION

UNIVERSITY OF LIMERICK OLLSCOIL LUIMNIGH

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No Lecture or Labs Next Week

We’re having open days on Thursday and Friday (15 & 16 Oct).

Assignment 1

- Release: Today at 5pm
- Deadline: Week 7 Thursday 22/10/2026
- Submission: TWO files, One PDF and One JAVA file
- Details: check Brightspace after 5pm!

The Goal Today Is to...

- Recap of past four lectures.
- Reflect and clarify any issues.
- Integrate knowledge gained over the past lectures.

Today, We Are Covering…

- Recap of Past Four Lectures
- Reflection & Discussion

Recap of Past Four Lectures

- Terminology
  - Concurrency vs Parallelism
- Identifying Parallel Algorithms
  - Characteristics of parallelisable problems
    - Decomposability, independence, and granularity.
  - Techniques for Identifying Parallelism
    - Decomposition analysis, data & Task parallelism
  - Amdahl's Law for measuring (theoretical) speedup
- Designing Parallel Algorithms
  - Foster's Design Methodology: Partitioning, Communication, Agglomeration, and Mapping
  - Common Patterns: Data Parallelism, Task Parallelism, Pipe-line Parallelism, and Divide & Conquer
- Common Faults in Parallel Programs
  - Race Conditions, Deadlock, Livelocks, and Starvation

Reflection & Discussion

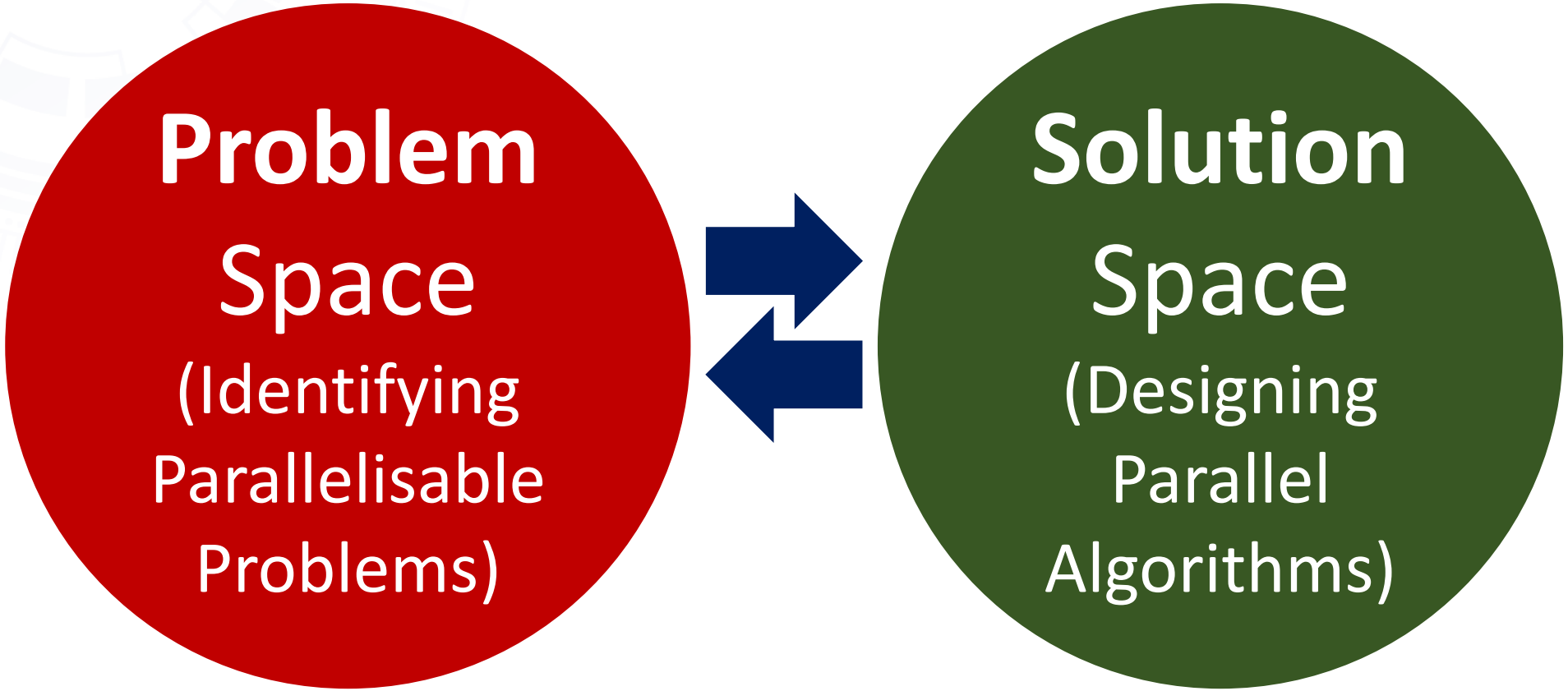
- How does the identification of parallelisable problems relate to the design of parallel algorithms?
- In what ways does the complexity of fault handling influence algorithm design choices?
- Based on your understanding of the lecture and lab material and your own expertise, what are some common mistakes developers make when trying to parallelise a problem? Why do these mistakes happen?

Reflection & Discussion

Is Every Problem Worth Parallelising?

Just because a problem *can* be parallelised, does that mean it should be!?

Reflection & Discussion



Case Study: Large-Scale Log Processing

A cloud-based application generates 100 million log records daily from multiple services. Each log record contains: Timestamp | ServiceID | Log Type | MSG | Response Time

- The system must perform the following operations:
  - Calculate the number of errors and warnings per service.
  - Compute average response times per service.
  - Detect unusual patterns, such as sudden increases in errors.
  - Generate hourly and daily summary reports.

- Identify independent and dependent tasks.
- Propose a suitable parallel algorithm.
- Identify synchronisation challenges.
- Explain how you would evaluate the proposed solution.

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What Are Your Questions?

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